

UNIT 2: RENEWABLE SOURCES OF ENERGY

Environment & Sustainability

Topics: Energy Management & Conservation | Hydrogen Energy | Solar Energy | Wind Energy | Tidal Energy | OTEC | Biomass | Geothermal | Energy Cycles | Carbon Emission & Sequestration | Green Engineering | Sustainable Urbanization & Sustainable Cities

1. Energy Management

Definition: Process by which a sector/organisation manages how much energy it produces, controls, monitors, and conserves, while generating enough to meet demand.

What is Energy Management?

- **Tracking & optimizing** energy consumption to conserve usage in a building.
- Manages energy production from **solar, nuclear, fossil fuel** and concerns optimal utilization at consumer devices.
- Enables owners/operators to:
 - **a) Reduce costs** — Energy = 25% of all operating costs in an office building.
 - **b) Reduce carbon emissions** — to meet sustainability goals and regulatory requirements.
 - **c) Reduce risk** — less energy demand = less vulnerability to price spikes or supply shortages.

Steps of Energy Management

MNEMONIC	<i>CIOER — Collect, Identify, Output, Execute, Repeat</i>
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1. Collecting and analysing continuous data.
2. Identify optimizations in equipment schedules, set points and flow rates to improve energy efficiency.
3. Calculate return on investment. Units of energy saved can be metered and calculated just like units of energy delivered.
4. Execute energy optimization solutions.
5. Repeat step two to continue optimizing energy efficiency.

2. Energy Conservation

Definition: The prevention of the wasteful use of energy, especially in order to ensure its continuing availability.

- Means **reducing consumption** by producing or using less energy.
- Achieved when **growth of energy consumption is reduced**, measured in physical terms.
- Can result from **productivity increase or technological progress**.
- **Energy Conservation vs. Energy Efficiency**: Separate but related concepts.
- Is the **deliberate practice** to save electricity, fuel oil, or gas to put it to additional use for additional productivity without spending extra resources or money.

3. New Renewable Energy Sources

Conventional/traditional sources are limited in quantity and cause pollution. Alternate (renewable/non-polluting) sources are abundant in nature and produce little to no pollution.

- Faster transportation, urbanization, industries — all need more energy.
- **Solar, wind, biomass** can meet a large extent of these requirements.
- Other sources like **tides, waves, geothermal** can also be used at suitable locations.

Types of Renewable Energy Sources

MNEMONIC	<i>SWiB-H-GT — Solar, Wind, Biomass, Hydro, Geothermal, Tidal</i>
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- Solar Energy
- Wind Energy
- Biomass / Bio-Energy
- Hydropower Energy
- Geothermal Energy
- Tidal / Ocean Energy
- Hydrogen Energy

4. Hydrogen Energy

Hydrogen fuel cells produce electricity by combining hydrogen and oxygen atoms. The reaction occurs across an electrochemical cell (like a battery) to produce electricity, water, and small amounts of heat.

- **Energy carrier**: Can store, move, and deliver energy produced from other sources.
- **Clean fuel**: When consumed in a fuel cell, produces ONLY water — zero harmful by-products.
- **Production sources**: Natural gas, nuclear power, biomass, solar, wind.
- **Applications**: Cars, houses, portable power, space exploration (fuels spaceships, aeroplanes, boats).

Advantages of Hydrogen Energy

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RCEEH — Renewable, Clean, Environment-safe, Efficient, Has wide applications

1. **Hydrogen is Renewable:** Cannot run out on a human timescale. Rich, abundant source of energy all around us.
2. **Hydrogen is a Clean Energy Source:** No harmful by-products released. Post-use, can even be converted to drinking water (NASA astronauts).
3. **Hydrogen Energy is Not Toxic:** Does NOT cause damage to human health unlike nuclear energy or natural gas.
4. **Hydrogen Energy is Highly Efficient:** 3 times more powerful than most fossil-based fuels, so LESS hydrogen is needed for the same tasks. Used in space exploration.

Disadvantages of Hydrogen Energy

MNEMONIC

VEDS — Volatile, Expensive, Difficult to store, Safety risk

1. **Hydrogen is Volatile:** Highly flammable and volatile — risky fuel to work with.
2. **Hydrogen is Expensive to Produce:** Steam-methane reforming and electrolysis are expensive. Mass production remains a challenge.
3. **Difficult to Store:** Lighter than gasoline — must be compressed into liquid and stored at low temperature. High pressure required.
4. **Hydrogen Can Be Dangerous:** Extremely flammable. No smell — sensors required to detect leaks.

Applications of Hydrogen Energy

- **Industrial processes:** Refining petroleum, treating metals, producing fertilizer, processing food.
- **Fuel cells (electricity):** Spacecraft, emergency power in buildings, remote locations not connected to power grids.
- **Vehicles:** Fuel cell vehicles (zero-emission). A fuel cell is 2-3x more efficient than a petrol engine.
- **Energy storage:** Stored as gas (salt caverns, aquifers) or liquid (cryogenic tanks at -253°C). NASA holds the two largest liquid hydrogen tanks.
- **Burning for electricity/heating:** Can blend with natural gas in combustion turbines (up to 100% hydrogen possible with modification).

5. Solar Energy

Solar radiation arriving at Earth's surface (called 'insolation') is in the form of electromagnetic waves with no mass. It is the most direct and pure form of energy.

- Intercepted by **collector systems** (flat plate or parabolic collectors). Heat energy used for heating air, water, or liquids.
- **Density of insolation** varies by altitude and meteorological conditions.
- A solar thermal device consists of: **collector, distribution system, storage, insulation.**
- For higher temperature and efficiency: **parabolic collectors** are used.

Harvesting / Harnessing Solar Energy

- **Solar Cells / PV Cells:** p-n junction diode working on photovoltaic technology. Direct conversion of solar radiation to DC electricity. Stored in batteries. Uses: calculators, watches, satellites, solar lamps, water pumping.
- **Solar Heat Collectors:** Used for heating buildings in cold countries. South-facing glass windows. Sunspaces with tile/brick floors absorb heat by day, release at night.
- **Solar Water Heaters, Solar Cookers, Solar Water Pumps**

Uses of Solar Energy

1. Air/water heating for industrial processes; solar pumps for water pumping.
2. Distillation and desalination of water (solar still).
3. Hot water for cooking, cleaning (solar water heater).
4. Electricity generation via photovoltaic cells.
5. Solar drying for food and vegetable products (cottage industries).
6. Solar cookers, lamps, lanterns, battery chargers for household purposes.
7. Solar cars/vehicles.

Advantages of Solar Energy

MNEMONIC	<i>Free RIND — Freely available, Renewable, Inexpensive long-term, No pollution, Domestic use</i>
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- Solar energy is freely available.
- Does not produce ash, fumes, or noise pollution — clean, noiseless, environment-friendly.
- Saves money in the long run.
- Renewable form of energy.
- Non-polluting. No wastes created by its use.
- Lots of domestic applications — advantageous for huge Indian population.

Disadvantages of Solar Energy

- **High initial cost.**
- **Energy storage:** Must be stored in batteries.
- **Space:** Large space needed for installation.
- **Intensity-dependent:** Energy generated depends on solar intensity.
- **Temperature limit:** High temperatures ($>40^{\circ}\text{C}$) cannot be achieved. Collector efficiency drops with rising temperature.
- **Weather-dependent:** Clouds affect amount and quality. May not be available when required.

6. Wind Energy

Wind energy is the kinetic energy of moving air. Uneven absorption of solar radiation by Earth's surface causes differences in temperature, density, and pressure — producing air movements.

- **Wind power** is a function of wind speed. Wind speed increases with height.

Method of Harnessing: Wind Mills

- Structure: **huge rotating blades mounted on a tower.**
- Blowing wind strikes blades, causing continuous rotation.
- Blades are linked to a **gear box and AC generator** — can drive water pumps, heaters, etc.

Advantages of Wind Energy

- The entire process of wind power is non-polluting.
- Wind energy is a sustainable source.
- Wind energy is renewable.
- Available at zero cost.
- Suitable for remote locations.

Disadvantages of Wind Energy

- **Intermittent:** Cannot be used as the sole resource of electricity. Requires backup/storage system.
- **Storage:** Storage technology is not fully developed.
- **Environmental damage:** Deforestation and loss of biodiversity.
- Birds may get killed.
- Noise pollution in local area; may affect TV reception.

Comparison: Solar vs Wind Energy

Solar Energy	Wind Energy
Can be extensively used anywhere.	Depends on geographical and meteorological factors.
Energy requirements can be met with small variations.	As wind speed and direction changes, energy production may fluctuate.
Can be used for solar heating, drying, electric generation, distillation, desalination.	Energy from wind cannot be used for such variety of uses.
Large number of applications — from houses to industries.	Applications are limited.
Depends on clouds (season), not available at night.	Does not depend much on clouds, season, time of day — continues to flow.
Installations do not require large open space (rooftop sufficient).	Huge open space required for wind mills.

7. Ocean Thermal Energy Conversion (OTEC)

OTEC uses the naturally occurring thermal gradient of the oceans. Warm surface water acts as heat source; cold deep water (~1000m) acts as heat sink. This thermal cycle generates electricity.

- **Minimum temperature difference required:** 20°C between heat source and heat sink.
- **Efficiency:** Very low (< 4%), but enormous potential merits investigation.

- **Key advantage over other renewables:** Can provide CONTINUOUS energy supply (unlike solar/wind).
- **Capacity factor:** 94% — making it one of the best power sources.
- **Zero emissions:** OTEC power plants produce no emissions.
- **24-hour power:** Powered by infinite solar energy; stored energy after sundown enables round-the-clock supply.

OTEC Technologies

- **Closed Cycle:** A working fluid (ammonia) is pumped through a heat exchanger for evaporation. Steam runs the turbine. Vapour condenses using cold deep water and returns to the heat exchanger.
- **Open Cycle:** Warm surface water is pressurized in a vacuum chamber and converted to steam which runs the turbine. Steam is condensed using cold ocean water from lower depths.

Advantages of OTEC System

MNEMONIC	<i>CRP-HET-F — Continuous, Renewable, Pollution-free, Hydrogen production, Enriches fishing, Tropical benefit, Floating plants</i>
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1. Power is continuous, renewable and pollution free.
2. Very little daily or seasonal variation in output (unlike other solar forms).
3. Drawing and returning sea water close to the thermocline — minimum environmental impact.
4. Electricity generated can be used to produce hydrogen.
5. Tropical/sub-tropical island sites can be freed from pollution caused by conventional fuels.
6. OTEC may enrich fishing grounds by bringing nutrients from deep unproductive waters to warmer surface waters.
7. A floating OTEC plant can generate power at mid-sea for offshore mining/processing of manganese nodules.

Limitations of OTEC System

- **Very high capital investment.**
- **Very low conversion efficiency (~3-4%)** due to small temperature difference between surface and deep water.
- Low efficiency + high capital & maintenance costs = **uneconomical for small plants.**

8. Tidal Energy

Tides are caused by the interaction of gravitational effects of the sun and moon with Earth's rotation. The relative motions of these bodies produce different tide cycles.

Understanding Tides

- **High Tide:** Moon/Earth gravitational field in a straight line — millions of gallons of water flow towards the shore.
- **Low Tide:** Moon/Earth fields perpendicular — water flows away from shore.
- **Spring Tides:** Moon perfectly aligned with Earth and Sun — much higher highs and lower lows. Occurs during full or new moon phase.

- **Neap Tides:** Gravitational pull of moon and sun oppose each other — smaller difference between high and low tides. Occurs during quarter moon phase.

Principle of Tidal Energy

- Tides created by **gravitational force of Earth with Moon and Sun**. Tidal energy is virtually inexhaustible → renewable.
- Tidal barrages/dams are built across narrow sea openings. Water rushes in during high tide → moves turbine blades → generates electricity.
- Tidal stream generators work like wind turbines — but water is **832 times denser than air** → higher power per unit.
- **World's first large-scale tidal energy station:** Rance Tidal Power Plant, France (1966).

Uses of Tidal Energy

1. **Tidal Electricity:** Most important use. Reliable as tides are predictable and uniform.
2. **Grain Mills:** Historically used for mechanical crushing of grains.
3. **Energy Storage:** Tidal barrages and reservoirs can be modified to store energy.
4. **Coastal Protection:** Tidal barrages prevent coastal damage during high storms; enable easy transport across estuaries.

Advantages of Tidal Energy

- Renewable form of energy.
- Does not require fuel.
- Does not produce ash and fumes — clean.
- Highly predictable (unlike wind/solar).
- Effective at low speeds — power can be generated at ~1 m/s water flow.
- Tidal turbines are 80% efficient — more than solar or wind.
- Vertical-axis turbines and coastal turbines are affordable to make with low environmental impact.

Disadvantages of Tidal Energy

- **Huge investment** required for construction.
- **Frequent equipment damage** is possible.
- **Environmental challenges:** Rotating blades dangerous to marine life.
- **Tidal Barrage:** May change the oceanfront, disrupting large ecosystems. Can affect turbidity and saltwater levels.
- **Tidal Lagoon:** Blade strike on fish, noise from turbines, changes in sedimentation.

India's Tidal Energy Potential

- **Tidal Energy:** ~12,455 MW potential. Key locations: Khambhat & Kutch regions, large backwaters.
- **Wave Energy:** ~40,000 MW theoretical potential.
- **OTEC:** ~1,80,000 MW theoretical potential in India.

9. Bio-Energy (Biomass Energy)

Definition: Bioenergy comes from any fuel derived from biomass — recently living organisms or their metabolic byproducts (e.g., cow manure, crop residues, municipal waste).

- **Unlike petroleum/coal:** Bioenergy is RENEWABLE.
- **Key point:** Carbon in biofuels was recently extracted from atmospheric CO₂ by growing plants → combustion does NOT result in a net increase of CO₂ in the atmosphere.
- **Note:** Like all energy generation methods, combustion of biomass does generate pollution as a by-product.

Types of Biomass Sources

- **Animal Waste:** Manure from pigs, chickens, cattle. Converted via anaerobic digestion into biogas.
- **Forestry Crops:** Fast-growing coppicing plants. Residues from thinning, sawdust, bark, woodchip.
- **Industrial Waste:** Solid (peelings, substandard food, coffee grounds) and Liquid waste (washing meats/fruits, wine making).
- **Municipal Waste:** Millions of tons in landfills. Converted by direct combustion or anaerobic digestion. Sewage — biogas production.

Forms of Biomass Energy

- **Biomass-Biopower:** Burning biomass directly, or converting it into fuel, to generate electricity.
- **Biomass-Biofuels:** Converting biomass into liquid fuels for transportation.

Problems with Biomass Energy

- Biomass has low energy density — transportation issue.
- Incomplete combustion — localized air pollution.
- Combustion itself produces pollution.
- Can lead to deforestation.
- Conflict of land use — competes with farming.
- Not fully competitive economically.
- Takes energy to make energy (energy input required).
- Increased need for pesticides and fertilizers.
- Waste products (wood, tires, sewage) can contain contaminants.

10. Geothermal Energy

Geothermal energy is the thermal energy generated and stored inside the Earth's crust. Earth's centre is at the same temperature as the Sun — nearly constant due to continuous nuclear fusion.

- High temperature and pressure → **rocks melt** → **mantle moves upward** → **molten rocks trapped in 'hot spots'**.
- When underground water contacts a hot spot → **steam is generated**.
- Hot water finding outlets at the surface = **hot springs**.

Harnessing Geothermal Energy

- A **hydrothermal convection system** is used: hole drilled deep underground → pipe inserted → steam trapped in rocks is routed to the surface.
- Steam is used to **turn turbine blades of an electric generator**.
- Alternative: steam heats water from external source → water rotates the turbine.

Types of Geothermal Power Plants

MNEMONIC	<i>DFB — Dry steam, Flash cycle, Binary cycle</i>
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- **Dry Steam Plants:** Oldest type. Use dry steam naturally produced underground. First built in Italy, 1904. Requires highest temperatures; least fluid flow. The Geysers, California (first drilled 1924) — largest geothermal electricity source.
- **Flash Cycle Steam Plants:** Most common type. Water must be $>180^{\circ}\text{C}$. Under its own pressure it flows up; pressure drop causes some water to 'flash' into steam which runs the turbine.
- **Binary Cycle Plants:** Expected most common type in future. Uses lower temperature water. Secondary loop with low boiling point fluid (pentane/butane). Heat exchanger transfers energy to this fluid, which vaporizes and drives turbine.

Applications of Geothermal Energy

1. **Electricity Generation:** Plants installed within 2-mile radius of geothermal reserve. Steam directly rotates turbines or generates steam from heated water.
2. **Farming:** In cold countries — heating greenhouses and irrigation water.
3. **Industry:** Food dehydration, milk pasteurizing, gold mining.
4. **Heating:** District heating systems — hot spring water transported via pipelines to buildings.

Advantages of Geothermal Energy

- **Renewable:** Free and abundant. Inexhaustible for ~4 billion years.
- **Green energy:** Non-polluting, environment-friendly. No harmful gases. No residue/by-products.
- **Employment generation:** Highly sophisticated plants → large-scale employment for skilled/unskilled workers.
- **Direct use:** Melting road ice, heating houses, greenhouses, public baths. Low maintenance cost after high initial cost.

Disadvantages of Geothermal Energy

- **Transportation & transmission:** Cannot be easily transported. Only efficient in nearby areas. Transmission risks emission of toxic gases.
- **High installation cost:** Huge investment in materials and human resources.
- **Intensive research required:** Sites can run out of steam due to temperature drop from excessive/irregular water supply.
- **Geographically limited:** Available in limited, often inaccessible regions (mountains, rocky terrain).
- **Environmental impact:** Drilling may release highly toxic gases near sites — sometimes fatal to workers.

11. Ocean Energy Resources

Ocean water movement creates kinetic energy in forms: wave energy, tidal energy, ocean current energy, salinity gradient energy, and ocean thermal gradient energy.

Types of Ocean Energy

- **Tidal Energy:** Power plants on river estuaries. India's potential: ~9,000 MW.
- **Wave Energy:** Generated by a device floating on the ocean surface or moored to the ocean floor.
- **Current Energy:** Underwater turbines (large propellers tethered to seabed) moved by marine currents.
- **Ocean Thermal Energy (OTEC):** Temperature difference between warm surface and cold deeper waters to generate steam and power.
- **Osmotic/Salinity Gradient Energy:** Energy from water movement across a membrane between saltwater and freshwater reservoirs.

Salient Features of Ocean Energy

- **Predictable & Reliable:** More predictable than wind. Endless flows = reliable future availability.
- **Global presence:** Tidal streams and ocean currents available almost everywhere.
- **Energy-rich:** Moving water is >800 times denser than air → multiplies kinetic energy by the same factor.
- **Unlimited usage area:** Vast and deep oceans eliminate competition for space.

Limitations of Ocean Energy

- **Deployment limited** in India; deployed technologies are underutilised.
- Most technologies at initial R&D, demonstration, or commercialization stage.
- **Marine environment uncertainty:** Corrosion from seawater salinity, offshore maintenance difficulties, environmental impact on landscapes and ecosystems, competition from fishing.

12. Energy Cycles

The energy cycle describes the interactions between energy sources within Earth's environment. Even small changes can lead to significant changes in long-term climate behaviour.

- Five key energy cycles in the ecosystem:
 - Energy Cycle
 - Water Cycle
 - Carbon Cycle
 - Oxygen Cycle
 - Nitrogen Cycle

The Energy Cycle

- Based on **flow of energy through the ecosystem**.
- Sunlight → **plants (producers)** convert to new plant material (flowers, fruits, trunks, roots) via photosynthesis.
- Energy depicted as **food pyramid / energy pyramid**: Plants (large base) → Herbivores (first-order consumers) → Humans (apex).

- Plants convert sunlight to **glucose (energy-rich sugar)**. Animals use glucose to produce ATP via mitochondria.

The Water Cycle

- Water cycles through **atmosphere, soil, rivers, lakes, oceans**.
- Rain → runoff into rivers/sea OR percolates into ground.
- Plants draw water from ground → transferred from leaves as **water vapour** → returns to atmosphere (transpiration).
- Key processes: evaporation, condensation, precipitation, runoff, infiltration, groundwater.

The Carbon Cycle

- Carbon is the basis of **all organic compounds** essential for life.
- Source: **CO₂ in air or dissolved in water**. Absorbed by plants via photosynthesis.
- Carbon returns to atmosphere when plants: **decay, are eaten/digested by animals, or burn in fires**.
- Maintains **oxygen levels and keeps CO₂ at acceptable levels**.
- Fossil fuels** account for >75% of global greenhouse gas emissions and ~90% of all CO₂ emissions.

The Oxygen Cycle

- Driven by photosynthesis** — responsible for the modern Earth atmosphere.
- Oxygen absorbed by plants and animals during respiration.
- Biogeochemical cycle of oxygen within four main reservoirs: **atmosphere, biosphere, hydrosphere, lithosphere**.
- Forestation programs are critical — plants play a crucial role in maintaining oxygen levels.

The Nitrogen Cycle

- Nitrogen-fixing bacteria and fungi** in soil give nitrogen to plants as nitrates.
- Nitrates → part of plant metabolism → form new plant proteins → consumed by herbivores → transferred to carnivores.
- Biogeochemical cycle: nitrogen converted into multiple chemical forms as it circulates among atmosphere, terrestrial and marine ecosystems.

13. Carbon Emission & Sequestration

Carbon Emission

- CO₂** makes up the vast majority of greenhouse gas emissions.
- Smaller amounts of **CH₄ (methane)** and **N₂O (nitrous oxide)** also emitted during fossil fuel combustion.
- Fossil fuels** (coal, oil, gas) account for >75% of global GHG emissions and ~90% of CO₂ emissions.
- CO₂ absorbs **infrared radiation** emitted from Earth → more CO₂ = more heat retained = **global warming**.

Carbon Sequestration

Definition: The long-term storage of carbon in plants, soils, geologic formations, and the ocean. Occurs naturally and through anthropogenic activities.

- Carbon is stored in: **vegetation, soil, detritus (~2,200 Gt globally), oceans (~920 Gt near surface).**
- Annual terrestrial sequestration: **~2.6 Gt.** Oceans absorb ~30% of human CO₂ emissions.
- Of all CO₂ emitted by humans, only **45% remains in atmosphere**; ~30% oceans, rest terrestrial.

Carbon Sources and Carbon Sinks

- **Carbon Sink:** Reservoir that retains carbon and prevents it from entering atmosphere. Examples: forests, oceans, limestone, underground fossil fuels.
- **Carbon Source:** Process that releases carbon into atmosphere. Examples: burning trees, volcanic activity, burning fossil fuels.
- **Deforestation** = carbon source. **Forest regrowth** = carbon sequestration (forests serve as carbon sinks).

Carbon Sequestration & Climate Change Mitigation

- Kyoto Protocol allows countries to receive **carbon sequestration credits** for land use, forestry activities.
- Activities: afforestation, reforestation, improved farming/forestry practices, revegetation.
- **IPCC:** Improved agricultural practices and forest-related mitigation can significantly remove CO₂ at low cost.
- Preserving **Amazon rainforests** important for continued carbon sequestration.

Carbon Capture and Storage (CCS)

- **Geoengineering proposal** for carbon sequestration.
- Process: CO₂ **separated from industrial emissions** → **compressed** → **transported** → **stored** in location isolated from atmosphere.
- Storage locations: **deep saline formations, depleted oil/gas reservoirs, deep ocean.**
- Can include 'scrubbing towers' and 'artificial trees' to remove CO₂ from surrounding air.

14. Green Engineering

Definition: The design, commercialization, and use of processes and products in a way that reduces pollution, promotes sustainability, and minimizes risk to human health and the environment without sacrificing economic viability and efficiency.

- Key insight: **Decisions made early (design and development phase)** have the greatest impact and cost-effectiveness on protecting health and environment.
- Objective: **Minimize overall environmental impact throughout entire product lifecycle** — from raw material extraction to ultimate disposal.

Four Approaches of Green Engineering

MNEMONIC	<i>W-M-P-P — Waste reduction, Materials management, Pollution prevention, Product enhancement</i>
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1. **Waste reduction**
2. **Materials management**
3. **Pollution prevention**
4. **Product enhancement**

Principles of Green Engineering

MNEMONIC	<i>UN-P-EDD-MER-HT — Use non-hazardous, Non-waste, Prevention, Easy sep, Durability, Design for need, Max efficiency, Renewable, Holistic, Think beyond</i>
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- Use of non-hazardous resources
- Prevention of waste
- Easy separation
- Durability of products
- Design for need
- Maximum efficiency
- Renewable rather than depleting
- Holistic approach
- Think beyond

Green Engineering Processes & Products

- Holistically use systems analysis and integrate environmental impact assessment tools.
- Conserve and improve natural ecosystems while protecting human health and well-being.
- Use life-cycle thinking in all engineering activities.
- Ensure all material and energy inputs/outputs are inherently safe and benign as possible.
- Minimize depletion of natural resources.
- Strive to prevent waste.
- Develop and apply engineering solutions while considering local geography, aspirations, and cultures.
- Create engineering solutions beyond current technologies — improve, innovate, invent to achieve sustainability.
- Actively engage communities and stakeholders in development of engineering solutions.

15. Sustainable Urbanization

Sustainability: Meeting the demands of the current population without compromising the demands of future populations. Sustainable living = meeting our needs now in a way that future people can also meet theirs.

Sustainable Urbanism: Study of cities and practices to build them, focusing on long-term viability by reducing consumption, waste, and harmful impacts while enhancing overall well-being.

Problems of Urbanization

MNEMONIC

Over-AS-WEPPSS-ALA — Overpopulation, Adequate housing, Supply of food, Water, Energy demand, Pollution, Solid waste, Sanitation, Altered land use

- Overpopulation
- Adequate housing
- Supply of food and other commodities
- Water availability and decreasing quality
- Energy demand
- Pollution
- Solid waste
- Sanitation and drainage facility
- Altered land use

Objectives of Sustainable Urban Projects

1. Conservation of identity, strengthening of neighbourhood and encouragement of cultural diversity and distinctiveness.
2. Expansion of public transport and its interconnection with existing and new developments.
3. Wise use of resources, minimising additional land take-up, and encouraging moderate urban density.
4. Safeguarding and interconnecting green spaces with networks, conservation of public spaces.
5. Assurance of social harmony and advancement of social and functional interaction.
6. Safeguarding existing jobs and creating new and innovative ones.
7. Advancing a culture of discourse.
8. Creating long-term partnerships between the community and the public/private sectors.
9. Participation in lifelong learning processes, seeing urban life in its wider context.

Benefits of Sustainable Urban Development

Environmental Benefits

- **Improved air quality:** Walking, biking, public transit cuts transportation emissions (>30% of PM2.5 pollution).
- **Smaller carbon footprint:** Better urban design reduces carbon pollution from transport (~25% of planet's energy-related GHG emissions).
- **Decreased car dependence:** Bike-sharing programs and dedicated lanes provide alternatives.

Economic Benefits

- **Lower cost for residents:** Energy-efficient travel, easier access to goods, lower parking costs.
- **Reduced congestion costs:** Mobility with density reduces traffic-related economic waste.
- **Higher property values:** Walkability and transit access increase real estate values globally.
- **Improved productivity:** Density boosts productivity and innovation through network effects.
- **Higher government revenues:** Smart development strategies cut capital, labour, and maintenance costs.

Social Benefits

- **Improved public health:** Mixed-use, transit-oriented development increases physical activity; reduces obesity and colon cancer risk.
- **Greater human mobility:** Saves time in traffic, expands economic and lifestyle choices.

- **Increased equality of access:** Better public transit, sidewalks, biking paths → greater access for those who cannot afford to drive.

16. Sustainable Cities

Definition: A city designed with consideration for social, economic, and environmental impact, providing a resilient habitat for existing populations without compromising the ability of future generations to experience the same. (UN SDG Goal 11)

- UN SDG Goal 11: Sustainable cities dedicated to **green sustainability, social sustainability, and economic sustainability**.
- A sustainable city creates a **permanent way of life across four domains:** ecology, economics, politics, and culture.

Major Problems of Indian Cities

- Urban Sprawl
- Overcrowding
- Housing
- Unemployment
- Slums and Squatter Settlements
- Transport
- Water
- Sewerage Problems
- Waste Disposal
- Urban Crimes
- Problem of Urban Pollution

Six Key Features of a Sustainable City

MNEMONIC

EV-PAW-GI — Electric Vehicles, Public resources, Access, Water, Green architecture, [get around without a car]

1. Make it easy to get around without a car.
2. Add Electric Vehicle (EV) charging stations.
3. Provide access to public resources and green spaces.
4. Improve water conservation and wastewater management.
5. Support urban farming.
6. Implement green architecture.

Technological Changes for Sustainable Cities

Industrial Processes

Procedures to aid manufacturing, usually carried out on a large scale, involving chemical, physical, electrical or mechanical steps.

Material Selection

- Identification of the design requirements

- Identification of criteria for material selection
- Evaluation of candidate materials
- Selection of materials

Pollution Prevention

- **Prevention at source** is fundamentally more desirable than recycling, treatment, and disposal.
- Can be done through **cost-effective changes in production and operation**.

Quick Reference: Advantages & Disadvantages Summary

All Energy Sources — At a Glance

Source	Key Advantages	Key Disadvantages
Hydrogen	Renewable, clean (only water by-product), non-toxic, 3x more efficient than fossil fuels	Volatile/flammable, expensive to produce, hard to store, no smell (sensors needed)
Solar	Freely available, clean, renewable, no pollution, domestic applications	High initial cost, battery storage needed, large space, cloud-dependent, temp limit >40°C
Wind	Non-polluting, sustainable, renewable, free, remote-area suitable	Intermittent, storage issues, deforestation, bird kills, noise pollution
OTEC	Continuous, renewable, pollution-free, little daily variation, enables hydrogen production	Very high capital, low efficiency (3-4%), uneconomical for small plants
Tidal	Renewable, no fuel, clean, predictable, 80% efficient turbines	Huge investment, frequent equipment damage, harmful to marine life, ecosystem disruption
Biomass	Renewable, carbon-neutral (no net CO2 increase), diverse feedstocks	Low energy density, combustion pollution, deforestation, land-use conflict
Geothermal	Renewable (~4B yrs), green, employment generation, direct use possible	Hard to transport, high installation cost, geographically limited, research-intensive, toxic gas risk

Key Numbers to Remember

Fact	Value
Energy in an office building (operating costs)	25%
OTEC minimum temperature difference required	20°C
OTEC efficiency	< 4% (3-4%)
OTEC capacity factor	94%
Hydrogen vs fossil fuel energy density	3x more powerful
Geothermal energy lifespan estimate	~4 billion years
Water density vs air density	>800 times denser
Tidal turbine efficiency	80%
CO2 of all emissions from fossil fuels	~90%
GHG from fossil fuels	>75% of global GHG

Total carbon in veg/soil/detritus globally	~2,200 Gt
Carbon sequestered annually by terrestrial ecosystems	~2.6 Gt
Oceanic carbon sink	~920 Gt
% human CO ₂ remaining in atmosphere	45%
India tidal energy potential	~12,455 MW
India wave energy potential	~40,000 MW
India OTEC potential	~1,80,000 MW
Rance Tidal Power Plant (first large-scale)	France, 1966
World's first geothermal power plant	Italy, 1904
Binary cycle geothermal working fluids	Pentane or Butane
Flash cycle: water temperature required	>180°C